

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech Examination (IT)****November 2019****IT473 Artificial Intelligence****Date: 13.11.2019, Wednesday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks:70****Instructions:**

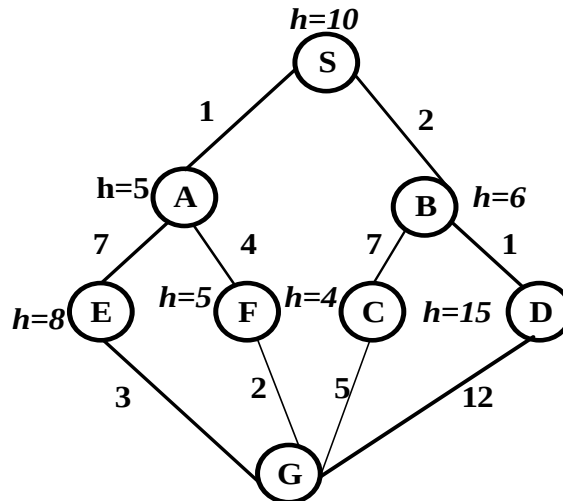
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

- Q - 1 Answer the following questions. [07]**
1. Match the following image formats to their correct number of channels **01**
 GrayScale
 RGB
 I = 1 channel, II = 2 channels, III = 3 channels, IV = 4 channels.
 (a) RGB -> I, GrayScale-> III (c) RGB -> III, GrayScale -> I
 (b) RGB -> IV, GrayScale-> II (d) RGB -> III, GrayScale -> II
 2. **True/False** : Fuzzy logic is extension of Crisp set with an extension of handling the concept of Partial Truth. **01**
 3. Which is used to compute the truth of any sentence? **01**
 (a) Semantics of propositional logic
 (b) Alpha-beta pruning
 (c) First-order logic
 (d) Both Semantics of propositional logic & Alpha-beta pruning
 4. Where does the bayes rule can be used? **01**
 (a) Solving queries (c) Decreasing complexity
 (b) Increasing complexity (d) Answering probabilistic query
 5. Define: Local maximum, Plateau, Ridge. **03**
- Q - 2.a Compare Breadth First Search and Depth First Search [04]**
- Q - 2.b Answer Following Questions. (Any Two) [10]**
1. Explain various steps of Natural Language Processing. **05**
 2. Consider the following sentences **05**
 If the unicorn is mythical, then it is immortal, but if it is not mythical, then it is a mortal mammal. If the unicorn is either immortal or a mammal, then it is horned. The unicorn is magical if it is horned.
 Convert the English into propositional logic implicative form and conjunctive normal form (CNF).
 3. Explain architecture and characteristics of Expert System. **05**

Q - 3 Answer Following Questions. (Any Two) [14]

1. What is Hill Climbing? Briefly explain disadvantages of hill climbing. Calculate good heuristic function and show the state space to solve 8-tile Puzzle problem using steepest ascent hill climbing method. 07
2. Show the contents of OPEN and CLOSED list for given graph using A* algorithm. 07



3. Define Cryptarithmic problem. Trace the constraints satisfaction procedure for solving the following cryptarithmic problem: 07

$$\begin{array}{r}
 \text{D O N A L D} \\
 + \text{ G E R A L D} \\
 \hline
 \text{R O B E R T}
 \end{array}$$

SECTION-II**Q - 4 Choose Most Appropriate Answer from the Following. [07]**

1. Relationship between vocabulary to represent different concepts is classified as 01
 - (a) dictionary
 - (b) glossary
 - (c) thesaurus
 - (d) all of above
2. Briefly explain Machine Learning? Define Supervised, unsupervised and reinforcement learning. 03
3. The task is to recognize English alphabetical characters (A, B, C, D, E, and F) in an image processing system. Define two fuzzy sets X and Y to represent the identification of characters A and B. 03

$$X = \{ 0.4/A + 0.3/B + 0.1/C + 0.1/D + 0.9/E + 0.8/F \}$$

$$Y = \{ 0.99/A + 0.8/B + 0.1/C + 0.2/D + 0.5/E + 0.5/F \}$$
 Find (i) union, (ii) intersection (iii) difference

[05]

Q - 5.a Consider the following axioms:

- John likes all kind of food.
- Apple and vegetable are food
- Anything anyone eats and not killed is food.
- Anil eats peanuts and still alive
- Harry eats everything that Anil eats.

Prove : "John likes peanuts." using Forward chaining approach.

Q - 5.b Discuss Bayesian network and its application. **[04]**

Q - 5.c What is knowledge representation? Explain Semantic network and Frame Representation methods to represent knowledge with suitable example. **[05]**

OR

Q - 5.b Consider the following axioms:

- All hounds howl at night.
- Anyone who has any cats will not have any mice. **[04]**
- Light sleepers do not have anything which howls at night.
- John has either a cat or a hound.
- (Conclusion) If John is a light sleeper, then John does not have any mice.

The conclusion you have to prove using Resolution

Q - 5.c Define monotonic and Non-monotonic Reasoning with suitable example. **[05]**

Q - 6 **Answer Following Questions. (Any Two)** **[14]**

1. How Artificial Neural network is generated based on biological Neural Network? Explain with suitable example. 07

2. **Consider the following sentences:** 07

- Every woman that can be burnt is a witch.
- Everything that is made of wood can be burnt.
- Everything that floats is made of wood.
- Everything that weighs the same as something that floats, does float too.
- This girl is a woman.
- This girl weighs the same as this duck.
- This duck floats.

i) Translate these sentences into formulas in predicate logic.

ii) Convert the formulas into clause form.

iii) Prove that "Girl is a Witch." using Resolution.

3. Decision tree algorithm falls under which learning category? Which kinds of problems can solve by it? Explain Decision tree algorithm with attribute selection measures. 07
